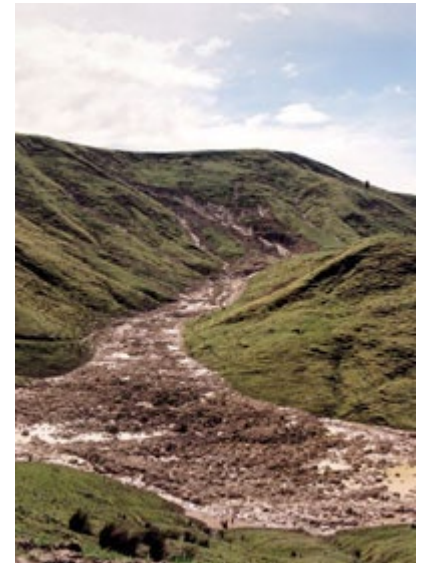


SEDIMENT TRANSPORT IN CHANNELS – SHEAR STRESS METHOD

ENCN442



Christchurch - after the storms...



Properties of Sediment

- Sieve size:

<i>Sieve size Diameter (d)</i>	<i>Material</i>
---- $< d < 0.004 \text{ mm}$	Clay
$0.004 < d < 0.062 \text{ mm}$	Silt
$0.062 < d < 2.0 \text{ mm}$	Sand
$2.0 < d < 64 \text{ mm}$	Gravel
$64 < d < \text{----}$	Cobbles and boulders

Diameter size (sieve and others)

- Diameter size is usually represented with a small “d” followed by a sub-script.
- d_p is the sieve opening which passes $p\%$ by weight. Thus d_p increases as p increases.
- d_{50} is the median.
- d_g refers to the geometric mean $= (d_{84.1} \times d_{15.9})^{1/2}$
- σ_g is the geometric standard deviation $= (d_{84.1} / d_{15.9})^{1/2}$
(84.1% and 15.9% are one standard deviation from median)

Diameter size (std)

- The geometric standard deviation is dimensionless and greater than or equal to 1.0 (uniform material).
 - Quasi uniform material: $1.0 < \sigma_g < 1.3$
 - Non-uniform material: $1.3 < \sigma_g$
- Some problems associated with σ_g are:
 - long tails on the distribution
 - bi-modal distributions: gravel bed rivers sometimes have sizes 2-8 mm underrepresented giving a frequency distribution with two peaks and a % finer distribution with a flatter section.
- d_{mm} = is just a diameter size of particles in millimetres

Phi index scale

d_{mm}	8	4	2	1	0.5	0.25	0.125	0.0625
φ	-3	-2	-1	0	1	2	3	4

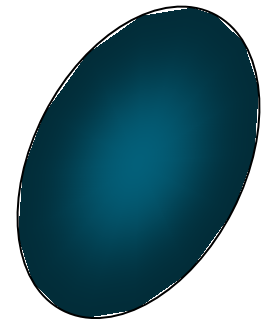
Example sediment sizes...

<i>River</i>	<i>location</i>	$d_{50\text{ (mm)}}$	σ_g
Missouri river	Omaha	0.20	1.20
Colorado river	Taylor Ferry	0.32	1.44
Waikato river	Karapiro	6.49	3.73
Crest River	Skew Bridge	19.2	1.85
Waimakariri	SH1 Bridge	28.0	2.6
Flood gate creek	Kaikoura	170	5.40

Shape Factor

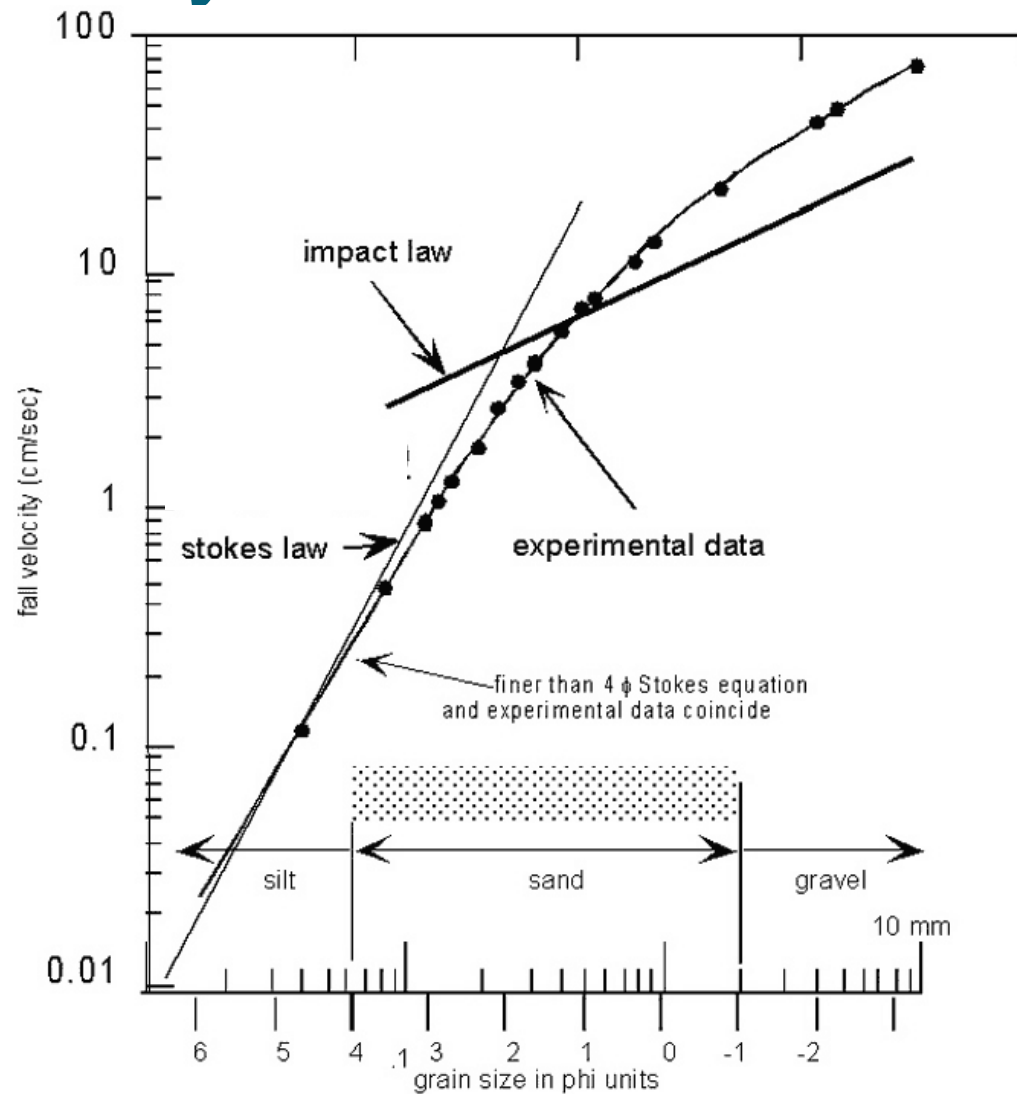
- The shape factor (S.F) is important in determining the behaviour of a particle in water for entrainment and fall velocity.
- Transportation rates may depend on the shape factor.

$$S.F. = \frac{c}{\sqrt{ab}}$$

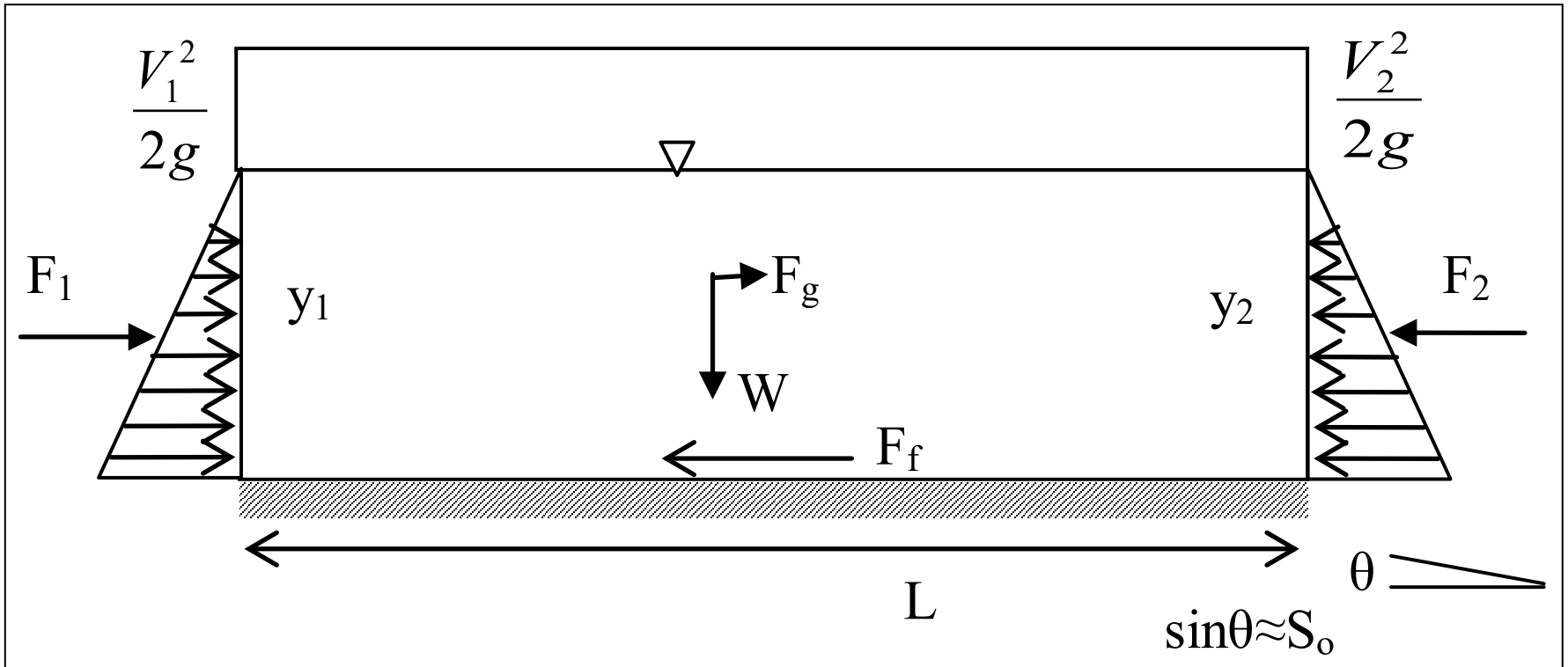


Canterbury stones?

Fall velocity..



Shear stress of flow



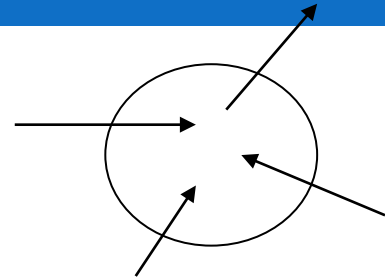
$$\tau_o = \gamma R S_o$$

- On board.....
- Derivation of boundary shear stress
- Derivation of Manning's n

Threshold conditions

- Refers to the flow conditions (discharge, velocity, or depth) that will initiate sediment motion of grain forming the boundary layer of the channel.
- Important in the design of a stable channel so no scouring will occur or so that no deposition will occur.
- Sometime both conditions are required.
 - Ex. irrigation channel coarse material will be used to protect the banks and bed and at the same time the channel will be designed so no deposition occurs.

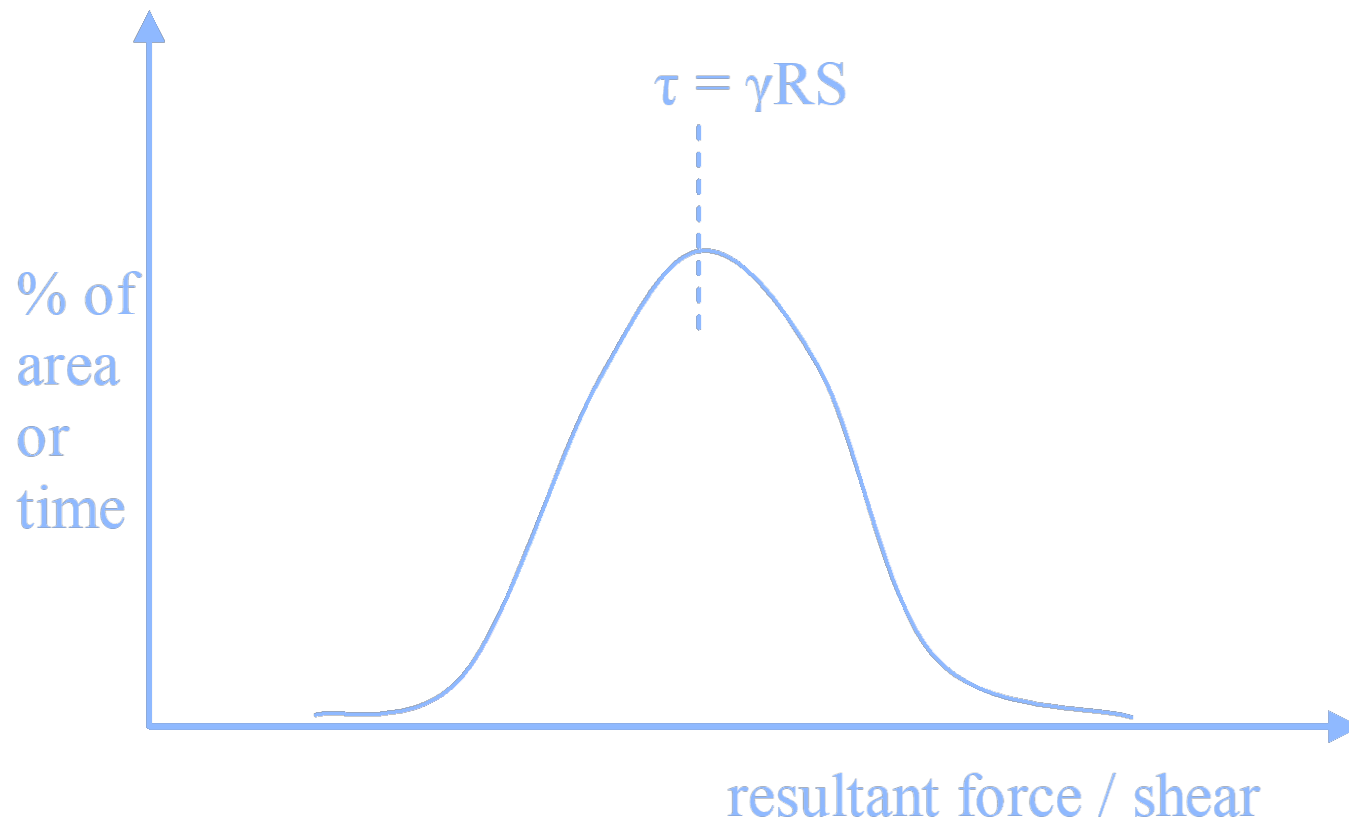
Forces on a grain



- Forces on a grain arise from drag and lift exerted on the grain from the fluid.
 - Drag – from skin friction (small component) and from pressure difference arising from flow separation.
 - Lift – from pressure differences between stagnation pressures in the bed and pressure due to flow and curved streamlines over the grain. Contributions also occur from vertical velocities in turbulent flow.
- Turbulent flow (Turbulent eddy's): will give rise to fluctuations in both lift and drag with time. Point of application and direction of forces will also vary with time.

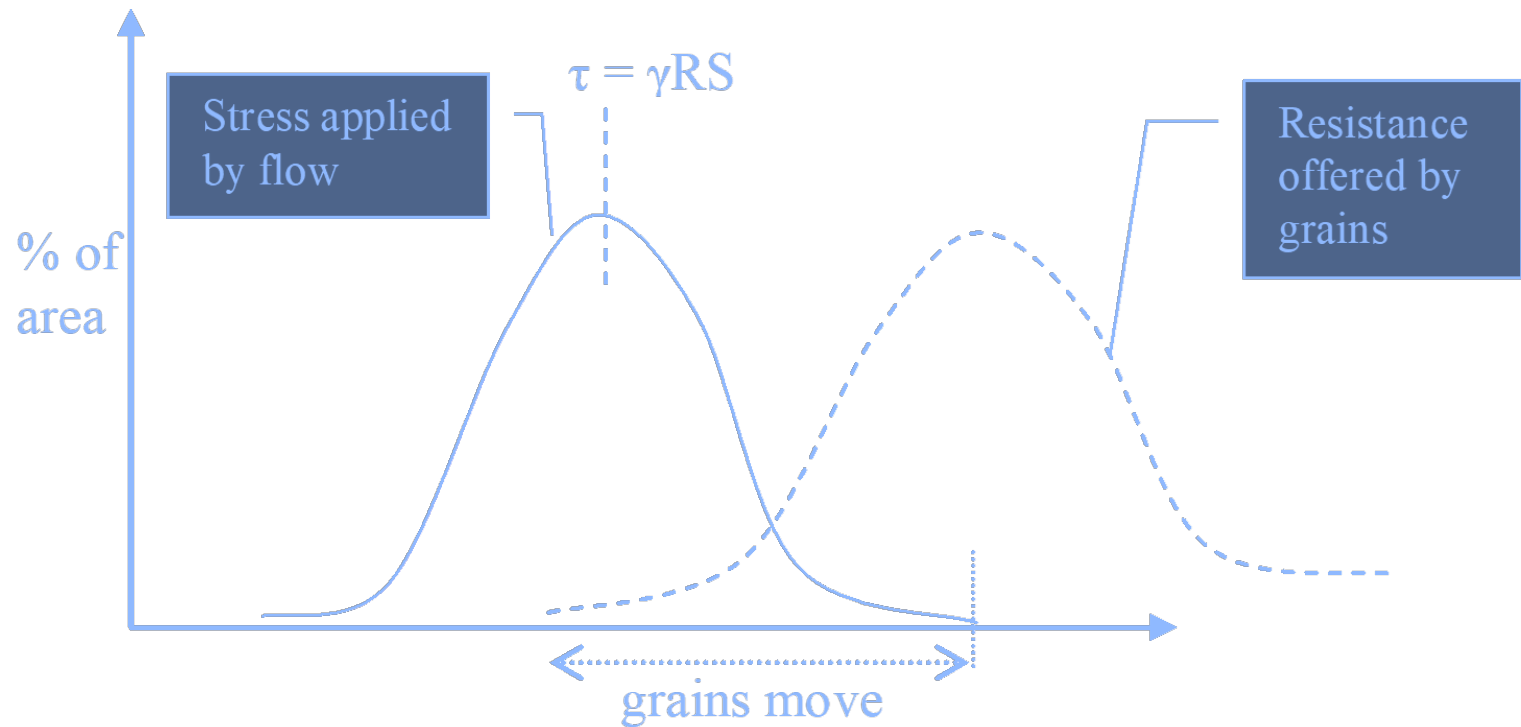
Threshold conditions

Particle shape, size, location, and orientation will also affect the forces on particular particles.



Resisting forces

- Resisting forces: comprise of the weight and any interference forces (mechanical or cohesive) between grains.



Critical conditions

- We can therefore define critical conditions as those for which an appreciable number of grains are in motion. “Appreciable” will depend on the particular application and will vary according to the amount of motion one can tolerate. This will depend on what you are designing.

Threshold (entrainment of particles)

Studies by White:

- White considered a particular grain and assumed it became unstable and moved by rotating about a downstream point of contact.
- Threshold conditions thus implied equality between resisting moment and overturning moment.

τ_c indicates critical or threshold conditions

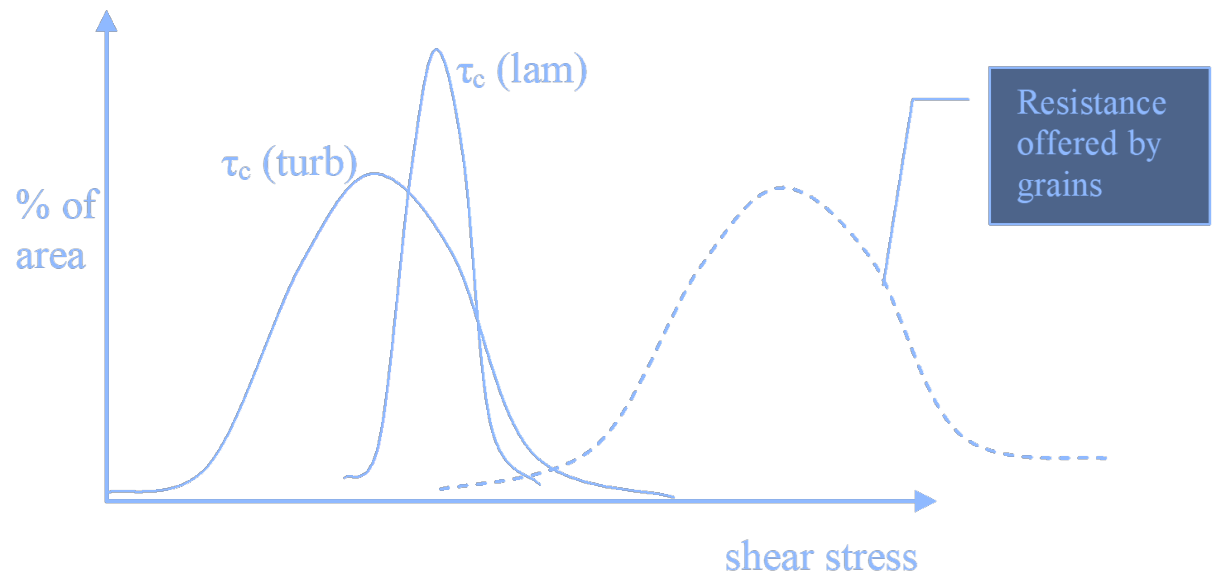
Critical shear stress (for entrainment) by White

- τ_c indicates critical or threshold conditions:

$$\tau_c = k(\gamma_s - \gamma)d \tan \phi$$

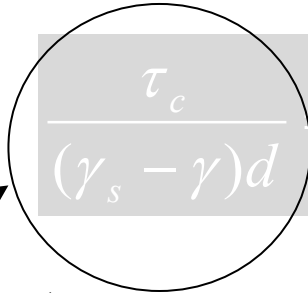
1. ϕ on average over the whole bed is equal to the angle of repose. When the bed is tilted to the angle of repose, the grains move at zero shear because the angle shown as ϕ has gone to zero.
2. White's results shows $\tau_c \propto d$. Submerged weight (W)
 $W \propto d^3$ and with
 $\tau_c \propto u_c^2$ then, $u_c^2 \propto W^{1/3}$ and $u_c^2 \propto W^{1/6}$

- Experiments with laminar flow showed White that $k=0.18$
- For turbulent flow k was only 50 to 60% of the laminar value.
- So in turbulent flow average threshold shear stresses are smaller. Measurements in turbulent flows show that shear stresses can reach 2 -2.5 times the mean value.



- So, as a result some grain motion will occur at lower values of mean shear although the shear stresses causing movement are the same in laminar and in turbulent flows. White considered the true value to be:

$$\tau_c = 0.18(\gamma_s - \gamma)d \tan \phi$$


$$\frac{\tau_c}{(\gamma_s - \gamma)d} = 0.18 \tan \phi$$

- The dimensionless number is known as the entrainment function and plays a very significant role in sediment transport.

Shields...

- The Shields diagram (Shields 1936) was developed from experimental measurements to predict the entrainment (incipient motion) of particles due to flow.
- The Shields diagram shows the functional relationship between the Boundary Reynolds number and the entrainment function.
- Shields (Shields 1936) argued that sediment grains would move when the drag force was equal to the weight resistance.
 - This is a sliding rather than rolling failure.

Limitations of Shields

- Limitations of Shields work are the following:
 - only flat beds were considered
 - threshold conditions were established by extrapolating sediment transport rate measurements back to zero
 - uniform grain sizes were used in all experiments
 - only fully developed turbulent flows were investigated
- In practice these conditions are not always met. A more serious limitation is that Shields always had R/d very large.
 - His results do not work for R/d less than 5 or for very steep slopes $S > 0.01$

Shields diagram

- See notes...

Allowable Shear Stress Method

- *Calculate Applied Shear Stress*

$$\tau_o = \gamma RS$$

Where τ_o = total bed shear stress, lbs/ft² or N/m²
 γ = specific weight of water, lbs/ft³ or N/m³
 R = hydraulic radius, ft or m
 S = energy slope, dimensionless

Keulegan (1938) presented a formula for fully rough flow, expressed as a function of the velocity and the ratio of hydraulic radius and boundary roughness

$$\tau = \frac{\rho V^2}{\left(\frac{1}{\kappa} \ln \frac{R}{k_s} + 6.25 \right)^2}$$

Where

- | | | |
|----------|---|---|
| V | = | depth-averaged velocity, ft/sec or m/sec |
| ρ | = | density of water, lbs-sec ² /ft ⁴ (slugs/ft ³) or kg/m ³ |
| κ | = | Von Karman's constant (usually taken to be 0.4) |
| k_s | = | roughness height, ft or m |

Actual shear stress values should be calculated for the banks, as well as the bed of a trapezoidal earth channel. Maximum stresses occur near the center of the bed and the lower third point of the banks.

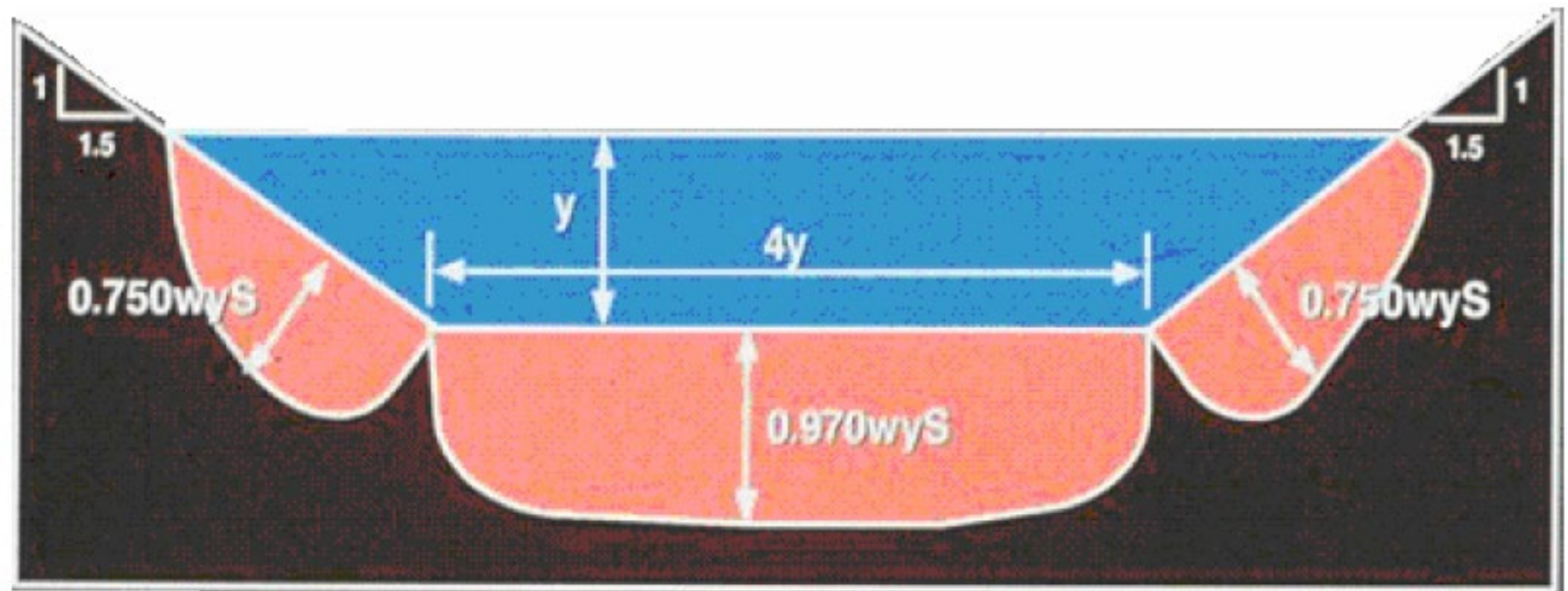


Figure 8-5. Lateral distribution of shear stress in a trapezoidal channel (Lane 1952). w = specific weight of water, y = depth and S = slope

Calculate Allowable Shear Stress

$$\tau_c = \tau^* (\gamma_s - \gamma) d$$

Where

τ^*	=	Shields Parameter, dimensionless
R^*	=	grain Reynolds Number = $u^* d / \nu$, dimensionless
τ_c	=	critical shear stress, lbs/ft ² or N/m ²
γ_s	=	specific weight of sediment, lbs/ft ³ or N/m ³
γ	=	specific weight of water, lbs/ft ³ or N/m ³
d	=	particle diameter, ft or m
u^*	=	shear velocity = $(gRS)^{1/2}$, ft/sec or m/sec
ν	=	kinematic viscosity of the fluid, ft ² /sec or m ² /sec
g	=	acceleration of gravity, ft/sec ² or m/sec ²

The Shields curve can be expressed as an equation, which is useful for computer programming and spreadsheet analysis.

$$\tau^* = 0.22 \beta + 0.06 \times 10^{-7.7 \beta}$$
$$\beta = \left(\frac{1}{\nu} \sqrt{\left(\frac{\gamma_s - \gamma}{\gamma} \right) g d^3} \right)^{-0.6}$$

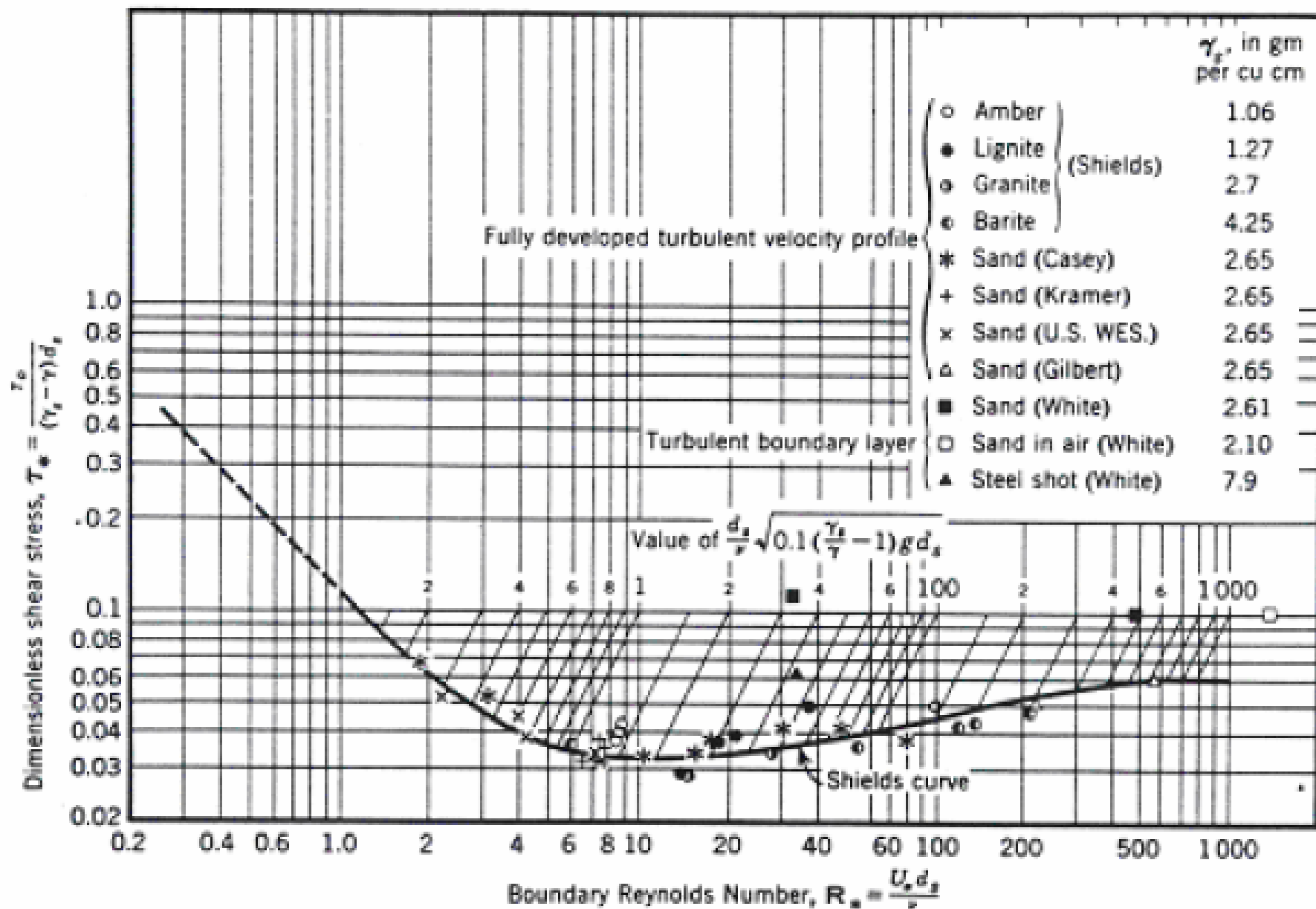


Figure 8-10. Shields diagram (from ASCE 1975)

Adjustment for Bed Forms

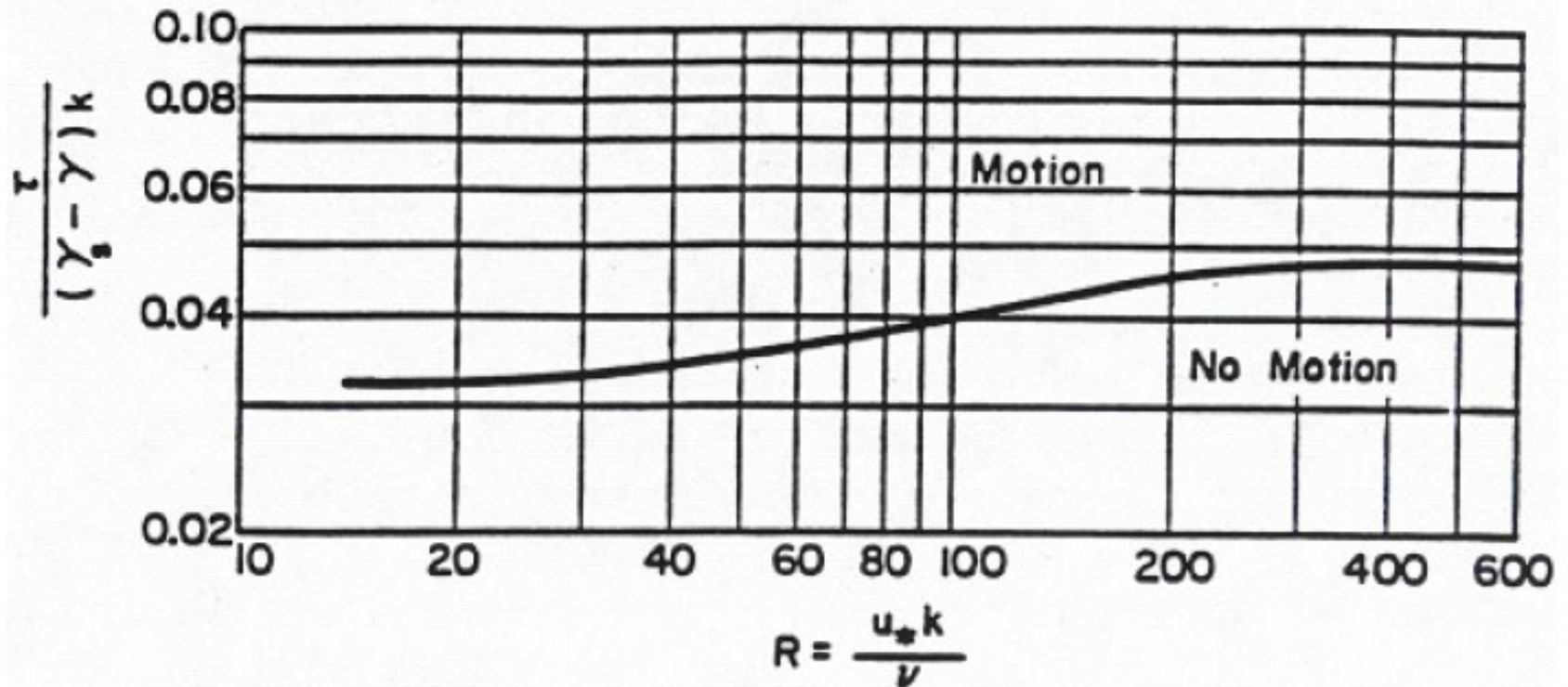


Figure 8-11. Gessler's reformulation of Shields diagram (Gessler 1971) τ is critical grain shear stress and k is grain diameter.

Procedure for Application of Allowable Shear Stress Method

Table 8-5. Characteristics of methods to determine allowable shear stress

	Theoretical Basis	Bed Characteristics	Safety Factor	Basis for Coefficients
Shields Parameter	The force on a single particle that initiates sediment motion	Non-cohesive	No	Flumes and channels
Gessler Probability	Probability distribution of force on a particle mixture	Non-cohesive	Yes	Flume
Lane	Force on surface area	Separate equations and charts for coarse and fine grained non-cohesive materials and cohesive material	Yes	Channels

Procedure for Application of Allowable Shear Stress Method

1. **Determine the hydraulics of the channel.** This includes the hydrologic determinations as well as the stage-discharge relationships for the channel being considered.
2. **Determine the soil properties of the bed and banks** of the design reach and of the channel upstream.
3. **Determine the concentration of the suspended sediment load entering the reach.** This is best accomplished by measurements. Channels with suspended sediment concentrations less than 1,000 ppm are considered *sediment free*. Sediment free flows are considered to have no effect on channel stability. Channels with suspended sediment concentrations greater than 20,000 ppm are considered to be *sediment laden*. Sediment laden flows are considered to enhance sediment stability by filling boundary interstices with cohesive material. If a significant portion of the inflowing sediment load is bed-material load it is likely that the channel is alluvial and threshold design methods are not applicable.

Procedure for Application of Allowable Shear Stress Method

4. **Check to see if the allowable shear stress approach is applicable.** Use Table 8-1.
5. **Compute the applied shear stress on the boundary of the channel being studied.** For non-cohesive bed materials grain shear stress can be calculated using the Limerinos equation or the Lane equation. If the Shields Parameter or Gessler Probability methods are used, calculate the grain shear stress using the Limerinos equation and the d_{84} of the boundary material. A safety factor should be added to this calculated grain shear stress if the Shields Parameter approach is to be used. If the tractive force method is used, calculate grain shear stress using Lane's equation with the d_{75} of the boundary material. Lane's equation already accounts for the safety factor so there is no need to increase the calculated applied shear stress. If the bed material is cohesive use the total shear stress as the applied shear stress. Use (with caution) Figures 8-8 and 8-9 to determine applied shear stress on the outside of bends.

Procedure for Application of Allowable Shear Stress Method

6. **Check the ability of the soil materials forming the channel boundary to resist the computed applied shear stress.** If the Shields Parameter method is used, determine an appropriate Shields Parameter and calculate critical shear stress. If the Gessler Probability method is used, calculate critical shear stress for each size class in the mixture using Figure 8-11. Then calculate the probability for each size class to stay using Figure 8-13. Finally, calculate the probability function for the bed mixture using Gessler's equation. For Lane's coarse grain soils method use Lane's equations with d_{75} to calculate allowable shear stress on the channel bottom and the channel slide slope. Lane's K factor for side slope allowable shear stress can also be adapted for use with the Shields Parameter and Gessler Probability methods. For Lane's method for fine grained soils use Figure 8-16 with the d_{50} and wash load sediment concentration to determine allowable shear stress. For cohesive materials, allowable shear stress should be determined by laboratory testing. Approximate values of allowable shear stress bases on soil properties can be determined from Figure 8-17.

Example Problem: Allowable Shear Stress Design

Given: A proposed flood channel has a bottom width of 8 ft, side slopes of 2/1 (H/V), and energy slope of 0.00085. The channel will flow at a normal depth of 4 ft, a velocity of 3.2 fps, and a discharge of 200 cfs. The soils are slightly angular sandy gravels, with D_{75} of 0.75 inches. Manning's coefficient for the entire channel is estimated at 0.025. The channel has a curve with radius of curvature of 40 ft.

Problem: Check stability using allowable shear stress approach.

Solution:

1. Calculate actual stresses on bed, sides and curve

a. Reference stress, τ_{∞}

$$\tau_{\infty} = \gamma D S_e (n_t / n)^2$$

$$n_t = (D_{75})^{1/6} \div 39 = (0.75)^{1/6} \div 39 = 0.0244$$

$$\tau_{\infty} = (62.4) (4) (0.00085) (0.0244 \div 0.025)^2 = 0.2021 \text{ psf}$$

b. Actual stress on channel bed, τ_b

$$\tau_b = \tau_{\infty} (\tau_b / \tau_{\infty})$$

Using Figure 8-6 $b/d = 8/4 = 2$ and $z = 2$:

$$\tau_b / \tau_{\infty} = 0.89$$

$$\tau_b = (0.2021) (0.89) = 0.1799$$

c. Actual stress on channel bed, curved reach, τ_{bc}

$$\tau_{bc} = \tau_b (\tau_{bc} / \tau_b)$$

Using Figure 8-8, $R_c/b = 40/8 = 5$:

$$\tau_{bc} / \tau_b = 1.56$$

$$\tau_{bc} = (0.1799) (1.56) = 0.281 \text{ psf}$$

d. Actual stress on channel sides, τ_s

$$\tau_s = \tau_\infty (\tau_s / \tau_\infty)$$

Using Figure 8-7, $b/d = 8/4 = 2$ and $z = 2$:

$$\tau_s / \tau_\infty = 0.76$$

$$\tau_s = (0.2021) (0.76) = 0.154 \text{ psf}$$

e. Actual stress on channel sides, curved reach, τ_{sc}

$$\tau_{sc} = \tau_s (\tau_{sc} / \tau_s)$$

$$\tau_{sc} = (0.154) (1.56) = 0.24 \text{ psf}$$

2. Calculate allowable stresses on beds and sides, τ_{Lb} and τ_{Ls}

a. Allowable stress on bed, τ_{Lb}

$$\tau_{Lb} = 0.4 D_{75}$$

$$= (0.4) (0.75) = 0.3 \text{ psf}$$

b. $\tau_{Ls} = 0.4 D_{75} K$

For K:

Use Figure 8-14 and $D_{75} = 0.75$ in:

$\Phi_R = 34.3^\circ$ for sub angular

Use Figure 8-15, $z = 2$ and $\Phi_R = 34.3^\circ$:

$K = 0.6$

$$\tau_{Ls} = 0.4 (0.75) (0.6) = 0.18 \text{ psf}$$

3. Compare actual with allowable stress for stability check

$$\tau_b = 0.1799, \tau_{bc} = 0.281 < \tau_{Lb} = 0.3 \text{ psf (OK)}$$

$$\tau_s = 0.154 < \tau_{Ls} = 0.18 \text{ psf} < \tau_{sc} = 0.24$$

Limitations and Cautions

- Aggradation of sediment inflow
 - Minimum velocity or shear stress that insures sediment transport (minimum velocity or nonsilting velocity)
 - Sometimes a “trickle channel” is used, a channel within a channel
- Bends and bank erosion
- Distinction between incipient motion and allowable velocity shear stress (incipient is less than allowable)
- Safety factor allowance for variability due to turbulence